

VID — VIRTUAL ID · TEAM MOBILIZATION BRIEF

Africa Ignite Hackathon 2026 · Prototype Phase

Prototype Phase: 24 April – 10 May 2026

SHORTLISTED · Building begins now

Status	SHORTLISTED — Prototype phase begins 24 April 2026
Submission deadline	10 May 2026, 18:25 UTC
Team size	5 members (max allowed)
Primary platform	Nokia Network-as-Code · CAMARA Open Gateway APIs
Repo structure	Single GitHub repo · everyone works on their own branch · team lead merges daily
Demo URL	Must be live on Vercel/Netlify by 8 May 2026 (2 days buffer before deadline)

1. What We Are Building — Read This First

VID (Virtual ID) is a Pan-African network-native identity system. We won our shortlisting with a specific, original idea. Every line of code we write in this prototype phase must serve that idea — not a different one, not a simpler one, not a more generic one.

The idea that got us shortlisted in one sentence

400 million Africans have no government ID but all have a SIM card. VID uses CAMARA telecom APIs as yes/no identity oracles — with full user consent and zero raw data storage — to issue a Pan-African network-verified identity certificate across all 54 AU member states.

The three things that make us different from every other team

- **Network-native:** Identity is derived from the mobile network itself — not from a document wallet or biometric scan. CAMARA APIs are the core, not a bolt-on.
- **Pan-African by design:** Each certificate speaks the user's country's ID language — Virtual NIN for Nigeria, Virtual Huduma Namba for Kenya, Virtual Ghana Card for Ghana. We cover 54 AU nations from day one via a JSON config file.
- **Privacy by design:** The network answers yes/no questions. No raw personal data is ever extracted, transferred, or stored. We only store a hash, a trust score, and a date.

2. Prototype Submission

The hackathon submission page lists an existing project as reference material — this is to demonstrate format and prototype quality expected, not to copy. Review it to understand the submission bar, then build something better.

Our submission for the hackathon is the VID (Virtual ID) Pan-African network-native identity prototype. This is the original project we are building and submitting, not a reference. All materials, code, and demos will showcase our VID system, which uses CAMARA telecom APIs to verify identity with full user consent and no raw data storage. The submission will demonstrate our unique approach—network-native verification, Pan-African coverage, and privacy by design—to meet and surpass the expected prototype quality and format.

Reference title	AR Spectro-scope
Reference idea	AR/VR app built on Unity platform
Video (reference)	To be updated (google drive link)
Presentation template	To be update (google drive link)
Demo (reference)	To be updated (loom link)
Source repo (ref)	https://github.com/jrcity/vid/
Run instructions (ref)	Markdown documentation on both frontend/backend are provided on each project

Note for Member 5 (QA/Storytelling): Download the presentation template from the link above. Our final slide deck must follow the same format. Study the demo video to understand pacing — our demo should be equally clear and under 3 minutes.

3. Shared Infrastructure — Set Up on Day 1

Before anyone writes a single line of feature code, the following must exist. Member 2 (Backend) owns setup. Team Lead confirms access for all.

GitHub repo	Create: github.com/[team]/vid-africa · Main branch is protected · Everyone makes feature branches · Team lead merges
Backend hosting	Railway.app (free tier) · Member 2 deploys the API server here · Share the URL with all members on Day 1
Frontend hosting	Vercel (free tier) · Member 4 deploys the React app here · Auto-deploys from GitHub main branch
Nokia NaC portal	Team lead registers at network.developer.nokia.com · Creates sandbox app · Shares API keys in team secure channel ONLY
API keys security	NEVER commit API keys to GitHub · Use .env files locally · Use Railway/Vercel environment variables in production
Daily standup	Every day at 19:00 Nigeria time · 5 minutes · Each person says: done today / doing tomorrow / any blockers
Progress tracking	Team lead maintains a shared Google Sheet: rows = tasks, columns = Not Started / In Progress / Done / Blocked

4. Role Assignments — Individual Task Briefs

Each role has Phase 1 tasks (complete by end of April 28 — first 4 days of prototype phase) and Phase 2 tasks (complete by May 7 — 3 days before deadline for final polish). Read your role card. Do not touch another member's area without checking with the team lead first.

Member 1 — Team Lead / Product (Redemption)

Role: Team lead, product owner, submission coordinator

Vision, pitch & final submission

Owns the idea integrity and final deliverables

Phase 1	Set up shared Google Sheet for task tracking — send link to all members today
	Register on Nokia Network-as-Code portal and create sandbox project — share API credentials securely
	Write the project README outline (GitHub) — sections: overview, how to run, API endpoints, screenshots
	Brief each team member individually using their section of this document
	Download the hackathon presentation template from the reference link and set up our slide structure
Phase 2	Write the demo script: 3-minute walkthrough showing enrollment → trust score → VID certificate → third-party QR scan
	Build the pitch slide deck following the hackathon template: problem, solution, APIs used, Pan-African scope, demo screenshots, team
	Coordinate final GitHub submission: check all branches merged, README complete, demo URL working, repo public
	Record or coordinate the recording of the final demo video (screen + voiceover following the demo script)
	Submit on HackerEarth before 10 May · Do not wait until the last day — submit on 9 May as a safety buffer
Output	Pitch deck (PPT) · Demo script · Final GitHub README · Submitted HackerEarth entry by 9 May

Member 2 — Backend Engineer

Role: API server, CAMARA integration, trust scoring engine

Node.js or Python · Must be comfortable with REST APIs

Owns the core engine that makes VID technically real

Phase 1	Read Nokia NaC CAMARA API documentation thoroughly · Confirm sandbox access works with a test call · Report to team lead by end of Day 2
----------------	--

	Set up Node.js (Express) or Python (FastAPI) project · Push to GitHub feature branch: feature/backend
	Deploy blank API server to Railway · Share the live URL in team channel
	Build the /health endpoint first — a simple GET that returns { status: 'ok' } — confirms deployment works
Phase 2	Build POST /enroll endpoint: accepts { phone_numbers: [], country_code: "", consent: true } · Returns trust score JSON
	Integrate SIM Swap API via Nokia NaC SDK · Map response to { simSwap: { stable: bool, lastSwapDays: number } }
	Integrate Number Verification API · KYC Match API · Location Verification API · Device Status API — each as its own service module
	Build the trust scoring engine: weighted algorithm — SIM Swap weight 35%, Number Verify 20%, KYC Match 20%, Location 15%, Device 10% · Output: score 0–100 + per-signal pass/warn/fail
	Build multi-SIM handler: accepts array of phone numbers, calls APIs for each, merges results, applies consistency bonus (+5 score if 2+ SIMs match same region and name)
	Build GET /verify/:vidId endpoint: returns { valid: bool, score: number, nationality: string, issueDate: string } — used by third-party QR scan · Never returns raw personal data
	Write Postman collection with test calls for all endpoints · Push to GitHub
Output	Live API server on Railway · All 5 CAMARA APIs integrated · Postman collection · Passing test calls demonstrated

Member 3 — AI / Data Engineer

Role: Country config, AI trust explainer, agentic monitoring layer

Python required · Basic knowledge of LLM APIs or prompt engineering is a plus

Owns the Pan-African intelligence that makes VID continental

Phase 1	Build country-config.json covering all 54 AU member states. Each entry must contain: country name, ISO code, phone prefix, VID type label (e.g. 'Virtual NIN'), AU region, and list of major MNOs. Start with the 10 priority nations from the submission document, then complete all 54.
	Build country-resolver.py: input = phone number → output = matching country config object. Test with +234, +254, +233, +27, +251 minimum.
	Document any countries where CAMARA API coverage is limited or unknown — flag these to team lead
Phase 2	Build the AI trust explainer module: input = trust score + signal breakdown JSON → output = 2–3 sentence plain English explanation of the result. Use Claude API or OpenAI API — keep prompts simple and output deterministic. Example output: 'Identity is strong. The SIM has been active on MTN Nigeria for over 2 years with no swap detected. A new device was registered 3 weeks ago which slightly reduces confidence.'
	Build the agentic monitor stub: a function that, given a VID ID and enrolled phone numbers, re-calls the SIM Swap API and returns { statusChange: bool, newStatus: string }. This simulates the real-time revocation agent — even a working stub counts for bonus points.

	Integrate the country resolver into the backend /enroll endpoint (coordinate with Member 2) — auto-populate nationality and VID type label from phone prefix
	Ensure the AI explainer output is accessible to Member 4 via the /enroll response JSON — add an 'explanation' field to the trust score response
Output	country-config.json (54 nations) · country-resolver.py · AI explainer module · Agentic monitor stub · All integrated with backend

Member 4 — Frontend Engineer

Role: Enrollment UI, trust score display, VID certificate, third-party verifier

React.js + Tailwind CSS · Mobile-first is mandatory — most African users are on phones

Owns everything the user and judge actually sees and touches

Phase 1	Set up React + Tailwind CSS project · Push to GitHub feature branch: feature/frontend
	Deploy blank React app to Vercel · Share the live URL in team channel
	Build the enrollment form screen: phone number input field, + Add another SIM button (up to 3), country auto-detection display (shows flag and country name from Member 3's resolver), consent checkbox with plain-language text, Submit button
	Connect enrollment form to Member 2's /enroll API endpoint (use the Railway URL) · Display a loading spinner while waiting
Phase 2	Build the trust score results screen: animated circular score ring (0–100), colour-coded (green 80+, amber 50–79, red below 50), signal breakdown table (one row per API: name, result, pass/warn/fail icon), AI explanation paragraph (from Member 3's explainer field)
	Build the VID certificate screen: full certificate card design (see design reference shared by team lead), nationality flag, country name, VID type label, holder name, masked phone number(s), trust score, signal summary, issue date, expiry date, unique VID ID, QR code (use qrcode.react library)
	Build the third-party verifier screen: text input for VID ID or QR scanner button (use react-qrcode library), calls /verify/:vidId endpoint, displays: valid/invalid status, nationality, trust score, issue date — nothing else (no raw personal data)
	Ensure all screens work on mobile (375px width minimum) · Test on Chrome mobile emulator · Responsive is non-negotiable
	Polish: loading states, error states (what happens if an API call fails), success animations on certificate issue
Output	Live demo URL on Vercel · All 4 screens working end-to-end · Mobile responsive · QR generation and scanning functional

Member 5 — QA / Research & Storytelling

Role: Testing, research, demo video, README, submission polish

No deep coding required · Strong writing and attention to detail essential

Owns the narrative that makes judges choose us

Phase 1	Download the hackathon presentation template (link in Section 2 of this brief) · Set up our branded slide deck with VID colors and structure · Share with team lead
	Write the impact research brief: for each of the 10 priority countries, find: approximate number of undocumented adults, what services they cannot access,

	and one real statistic or news story. Keep each country to 3 bullet points. Sources: World Bank, GSMA Intelligence, African Development Bank.
	Write the competitive landscape summary: list 5 existing identity apps or solutions in Africa · For each write one sentence on what they do and one sentence on why VID is different
	Set up 10 test scenarios (see list below) and test each against the live demo URL once Member 4 has it deployed
Phase 2	Run all 10 test scenarios and log results in the tracking sheet. Flag any broken flows to the relevant team member with a clear description of: what you did, what you expected, what actually happened.
	Write the GitHub README: Project overview (2 paragraphs), Problem and solution (2 paragraphs), How to run locally (step by step), API endpoints table (endpoint / method / input / output), Screenshots of all 4 screens, Team member roles
	Record the final 3-minute demo video: use Loom or OBS · Screen record the live demo URL · Use the demo script written by Member 1 as your voiceover guide · Keep it calm, clear, and professional — no background noise
	Proofread the entire submission: HackerEarth description text, slide deck, README, demo video captions if any · Fix any spelling, grammar, or factual errors before team lead submits
Output	Tested all 10 scenarios · GitHub README complete · 3-minute demo video · Polished slide deck · Submission proofread and signed off

5. Test Scenarios — for All Members

Test each of these scenarios on the live demo URL and log pass/fail in the tracking sheet:

Scenario 1	New Nigerian user enrolls with 1 SIM — no recent swap — expect high score, Virtual NIN certificate
Scenario 2	New Kenyan user enrolls with 2 SIMs — both stable — expect high score, Virtual Huduma Namba, multi-SIM bonus applied
Scenario 3	South African user — SIM swapped 5 days ago — expect lower score, SIM Swap signal flagged as warn
Scenario 4	Ethiopian user — only 1 SIM, KYC partial match — expect moderate score, explanation reflects partial KYC
Scenario 5	User enters invalid phone number format — expect friendly error message, no crash
Scenario 6	User does not check consent checkbox — expect form to refuse submission with explanation
Scenario 7	User enrolls, certificate issued, third-party scans QR code — expect verifier screen shows valid status and nationality only
Scenario 8	Third-party enters invalid VID ID — expect verifier screen shows 'ID not found' cleanly
Scenario 9	Mobile test: run full enrollment flow on 375px viewport (Chrome dev tools) — expect no broken layouts or overflowing text
Scenario 10	Ghana user with 3 SIMs all matching — expect highest score in test suite, all signals green, AI explanation congratulatory tone

6. Build Timeline — 11 Days

Days 1–2 (Apr 29–30)	SETUP: GitHub repo · Railway deployment · Vercel deployment · Nokia NaC sandbox confirmed · Tracking sheet live · All members briefed
Days 3–5 (Apr 30)	CORE: Backend /enroll endpoint · 3 CAMARA APIs integrated · Country config built · Enrollment form connected to API
Days 6–9 (May 1–May 3)	BUILD: All 5 CAMARA APIs integrated · Trust scoring engine complete · All 4 frontend screens functional · AI explainer integrated
Days 10–12 (May 3–5)	INTEGRATION: Full end-to-end flow working · QR generation and scanning · Multi-SIM handler tested · Member 5 runs all 10 test scenarios
Days 13–14 (May 6–7)	POLISH: Bug fixes from test scenarios · Mobile responsiveness final checks · Slide deck final version · Demo video recorded
Day 15 (May 8)	BUFFER: Final review · README complete · All submission materials ready · Team lead does submission dry-run
Day 16 (May 9)	SUBMIT: Team lead submits on HackerEarth before 18:00 UTC · One day early as safety buffer